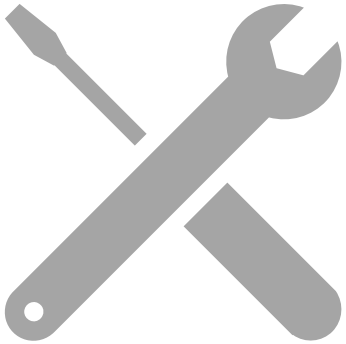
Predicting the Conditions of Water wells in Tanzania

By Mercy Apondi

Problem Statement

Many water wells become non functional over time., we are looking to find out why using Machine Learning



We need a system to predict well condition to prioritise maintenance.

Business Understanding

- Tanzania faces challenges in maintaining water infrastructure. Many wells fail over time, and resources are limited.
  This project aims to help decision-makers identify which wells need attention most urgently:
  - Limited access to clean water
  - Many wells are non functional or deteriorating
  - Need for efficient resource allocation

Data Understanding

THE DATASET CONTAINS DETAILED INFORMATION ABOUT WELLS, INCLUDING LOCATION, CONSTRUCTION, AND OPERATIONAL FACTORS.
THE TARGET VARIABLE CLASSIFIES WELLS INTO THREE CATEGORIES BASED ON FUNCTIONALITY.

59,400 WATER WELLS

FEATURES: LOCATION, PUMP&WATER CHARACTETRISTICS ,CONSTRUCTION DETAILS ETC

PREDICTIVE VARIABLE: FUNCTIONAL, NEEDS REPAIR OR NON FUNCTIONAL PUMPS.

Data Pre-Processing

Handling of Missing values.

This step involves handling missing values, correcting inconsistencies, and transforming raw data into a format suitable for modeling.

Feature Engineering

Informative features are created or existing ones are transformed to improve model performance ensuring the model captures patterns that may eventually influence the water well's operational status.

Use of Pipelines

Ensure Consistent Pre Processing

Modeling Approach

Baseline; Logistic Regression

Advanced: Random Forest

Ensemble; Voting Classifier

Model Evaluation using Accuracy & Confusion Matrix

This provides a clear view of functional and non functional wells using the different model performances which are measured using accuracy, which reflects the proportion of correct predictions.

Evaluation Metrics

- Accuracy
- Precision, Recall, F1-score
- Confusion Matrix

Key Insights

The model revealed that geography plays a major role in well performance. Additionally, older wells and those with low water quantity are more likely to fail.

Location is the strongest predictor

Older wells fail more often

Water quantity strongly affects functionality

High population increases strain

Recommendations

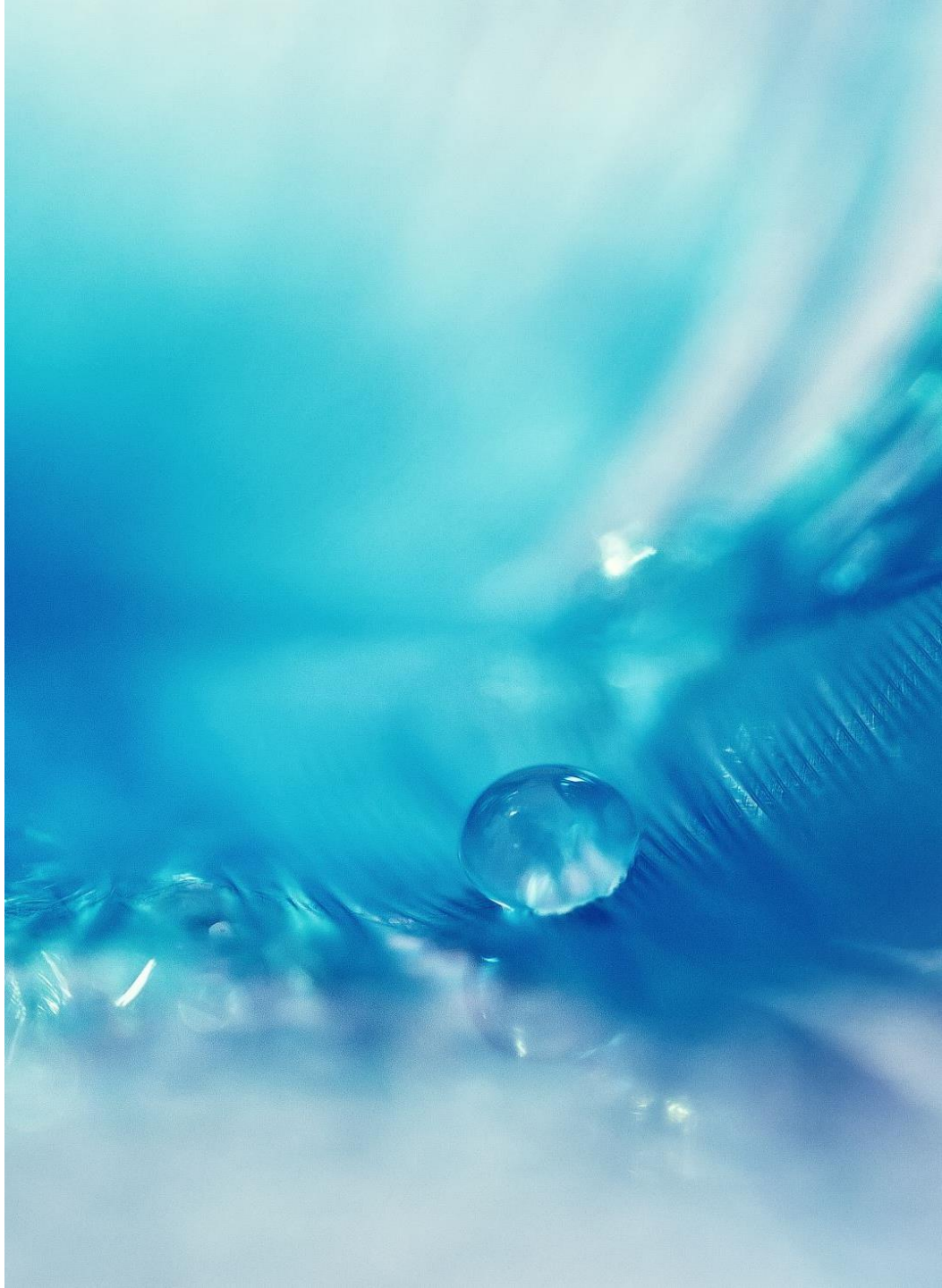
- Use ensemble model for predictions

- Improve data collection

Prioritize repairs for wells predicted as non-functional as the older wells have declining water quantity.

Conclusion

- This project demonstrates the application of machine learning to a real-world infrastructure problem.
- By leveraging data-driven insights, it is possible to identify failing water wells and support better planning and resource allocation, ultimately improving water access in Tanzania.



Project’s Next Steps

Future improvements could focus on better identifying wells that are close to failure. This model could also be deployed as a real-time decision-support system.

Improve detection of ‘needs repair’ wells

Perform geospatial analysis

Tune models further

Deploy as a decision tool

Thankyou.

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